

**SIMATS ENGINEERING**  
**C01-ASSESSMENT 3**  
**CSA0310-DATA STRUCTURE**  
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**C Program to output Prediction**← Back

**QUESTIONS**

**Q1**

Reverse a list

1 mark

✓

**Q2**

Reverse a string

1 mark

✓

**Q3**

Searching

1 mark

✓

3/3 answered

**Q1 Reverse a list**

1 mark

Write a C program that takes an array as input and returns it reversed.

**Your Code**

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i;
5     int arr[100];
6
7     scanf("%d", &n);
8
9     for (i = 0; i < n; i++) {
10        scanf("%d", &arr[i]);
11    }
12
13    for (i = n - 1; i >= 0; i--) {
14        printf("%d ", arr[i]);
15    }
16
17    return 0;
18 }
```

**Input**

5  
10 20 30 40 50

**Output**

50 40 30 20 10

Submitted

**Q2**

Reverse a string

1 mark

✓

**Q3**

Searching

1 mark

✓

3/3 answered

**Your Code**

```
1 #include <stdio.h>
2
3 int main() {
4     int arr[100], n, i, flag = 1;
5
6     // Read the size of the array
7     printf("Enter the number of elements: ");
8     scanf("%d", &n);
9
10    // Read array elements
11    printf("Enter %d elements:\n", n);
12    for (i = 0; i < n; i++) {
13        scanf("%d", &arr[i]);
14    }
15
16    // Check if the array is a palindrome
17    for (i = 0; i < n / 2; i++) {
18        if (arr[i] != arr[n - i - 1]) {
19            flag = 0;
20            break;
21        }
22    }
23
24    if (flag)
25        printf("The array is a Palindrome.\n");
26    else
27        printf("The array is Not a Palindrome.\n");
28
29    return 0;
30 }
```

**Input**

5  
1 2 3 2 1

**Output**

Enter the number of elements: Enter  
5 elements:  
The array is a Palindrome.

1 mark

Q3

Searching

1 mark

3/3 answered

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int arr[100], n, key, low, high, mid, i;
6
7     scanf("%d", &n);
8
9     for(i = 0; i < n; i++)
10         scanf("%d", &arr[i]);
11
12     scanf("%d", &key);
13
14     low = 0;
15     high = n - 1;
16
17     while(low <= high)
18     {
19         mid = (low + high) / 2;
20
21         if(arr[mid] == key)
22         {
23             printf("Element found at position %d", mid + 1);
24             return 0;
25         }
26         else if(key < arr[mid])
27             high = mid - 1;
28         else
29             low = mid + 1;
30     }
31
32     printf("Element not found");
33
34     return 0;
35 }
```

5

10 15 20 25 30

15

Output

Element found at position 2